

Are projectors invertible? Are they reducible?

Solution

Let $\mathcal{H}$ be a Hilbert space and let $P : \mathcal{H} \rightarrow \mathcal{H}$ be a linear projector. By definition,

$$P^2 = P.$$

If P is an orthogonal projector, then additionally $P^\dagger = P$, but the arguments below apply to all projectors.

Invertibility of projectors

Step 1: Necessary condition for invertibility

Assume that P is invertible. Then there exists an operator P^{-1} such that

$$P^{-1}P = PP^{-1} = I.$$

Multiplying the identity $P^2 = P$ on the left by P^{-1} gives

$$P = I.$$

Conclusion

The only invertible projector is the identity operator. Therefore:

- $P = I$ is invertible,
- Any nontrivial projector ($P \neq 0, I$) is *not* invertible.

In particular, orthogonal projectors are not invertible unless they project onto the entire Hilbert space.

Reducibility of projectors

Step 1: Definition of reducibility

An operator T on $\mathcal{H}$ is said to be *reducible* if there exists a nontrivial closed subspace $\mathcal{M} \subset \mathcal{H}$ such that

$$T(\mathcal{M}) \subset \mathcal{M}.$$

Step 2: Natural invariant decomposition

For any projector P , the Hilbert space decomposes as

$$\mathcal{H} = \text{Ran}(P) \oplus \text{Ker}(P).$$

Both subspaces are invariant under P :

$$\begin{aligned}P(\text{Ran}(P)) &\subset \text{Ran}(P), \\ P(\text{Ker}(P)) &= \{0\} \subset \text{Ker}(P).\end{aligned}$$

Conclusion

Unless $P = 0$ or $P = I$, both $\text{Ran}(P)$ and $\text{Ker}(P)$ are nontrivial invariant subspaces. Hence,

Every nontrivial projector is reducible.

Summary

- Projectors are invertible if and only if $P = I$.
- Nontrivial projectors are never invertible due to nonzero kernels.
- Every nontrivial projector is reducible.

Insights

- Non-invertibility reflects information loss in projection (important in quantum measurements).
- Reducibility arises naturally from the decomposition into measured and unmeasured subspaces.
- Projectors provide canonical examples of operators with invariant subspaces.

MCQs

1. Which projector is invertible?

- (a) $P = 0$
- (b) $P = I$
- (c) Any orthogonal projector
- (d) None

Answer: (b)

2. Why is a nontrivial projector not invertible?

- (a) It is unbounded
- (b) It has a nontrivial kernel
- (c) It is not linear
- (d) It is not selfadjoint

Answer: (b)

3. Why are projectors reducible?

- (a) They commute with all operators
- (b) $\mathcal{H} = \text{Ran}(P) \oplus \text{Ker}(P)$
- (c) They are unitary
- (d) They are nilpotent

Answer: (b)